

INTRODUCTION TO IOT & ITS TOOLS

What is IOT?

Ts. Dr. Razif bin Razali

<https://github.com/razifrazali66/IoT.git>

DEFINITION AND BASIC CONCEPTS OF IOT

"A network of physical objects (things) embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet."

THINGS/DEVICES

CONNECTIVITY

UI/APPLICATIONS

DATA COLLECTION
& PROCESSING

ANALYTICS & INTELLIGENCE

SECURITY AND PRIVACY

REAL-WORLD EXAMPLES & APPLICATIONS

SIMPLEST	MODERATE	HIGHLY COMPLEX
Consumer IOT, smart lighting, wearable fitness, smart plug, smart thermostats, trackers	Smart Home Security Systems, Smart Kitchen Appliances, Connected Cars, Smart Retail	Industrial IOT (IIOT) for predictive maintenance, Smart City Traffic Management, Smart Farming, Smart Grids, Internet of Medical Things (IoMT).

THE IMPORTANT OF IOT & ITS FUTURE PROSPECTS



Jobs ▾ | **Date posted** ▾ | **Experience level** ▾ | **Company** ▾

iot in Malaysia
346 results

Set alert ☐



Junior IoT Engineer

Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia · 1 week ago · 70 people clicked apply

Responses managed off LinkedIn



IoT Data Engineer

Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia · 5 days ago · **11 people clicked apply**

Responses managed off LinkedIn

Remote

Full-time

indeed

Find jobs

IOT Executive/ IT Agritechnology Executive

Kualu Lumpur Kepong Berhad  · 4.1 ★

Ipoh

Apply now 



Farm AI/ IoT Engineer

Sustainable Harvest Sdn Bhd  | Puchong | RM 1,900.93 - RM 4,145.07 a month

Apply now



400+ jobs ?

Key Responsibilities

- **Design, Installation and Maintenance:**
- Has a capabilities to design, assist in the installation, configuration, and maintenance of IoT-based farming equipment, including sensors, weather stations, and control panels.
- A.I Computer Vision and with AWS environment is an advantage.
- Perform basic troubleshooting and repairs on hardware and software components.
- **Project Execution:**
- Support the deployment of smart farming solutions at various project sites.
- Ensure the quality and reliability of installed systems in accordance with project specifications.

Embedded Systems Engineer (IoT Engineer/ Automation Engineer)

Elitez Pte Ltd  Kuala Lumpur | Up to RM 10,000 a month

Apply now



Job Requirements

- Bachelor's degree in Computer Science, Electrical Engineering, or a related field.
- Minimum of 2 years of experience in designing and developing IoT devices.
- Strong programming skills in C, C++, and Python.
- Hands-on experience with microcontrollers and embedded systems such as Raspberry Pi, Arduino, ESP32, or STM32.
- Familiarity with IoT communication protocols and cloud platforms like AWS IoT, Azure IoT, or Google Cloud IoT.
- Solid understanding of IoT security measures and best practices.
- Strong problem-solving abilities and the capacity to work in a dynamic, fast-paced environment.
- Willingness to travel frequently as part of the role.

Job Type: Full-time

Pay: Up to RM10,000.00 per month

Schedule:

- Day shift

Work Location: On the road



IoT Engineer
Senior Industrial IoT Engineer
IoT Full Stack Engineer (Junior Level)
Embedded Software Engineer (RTOS/Firmware)
IoT Market Intelligence Analyst
Technical Support Customer Service (IoT & After-Sales) / IT
Executive / IT Support Engineer
Internship (Various IoT Fields)
PLC Integration Systems with IoT Program
Business Admin Executive (IoT, Tech, Smart Vending)

INTRODUCTION TO IOT & ITS TOOLS

Introduction to ESP8266

Ts. Dr. Razif bin Razali

WHY ESP8266 POPULAR FOR IOT PROJECTS

Cost-effective:

ESP8266 modules are very affordable, making them accessible for a wide range of projects.

Low-power consumption:

The module is designed for efficient power usage, which is crucial for battery-powered IoT devices.

Integrated Wi-Fi:

The built-in Wi-Fi allows for easy network connection and data transmission.

Microcontroller capabilities:

It can process data, interact with sensors, and execute custom applications, offering flexibility beyond just Wi-Fi functionality.

Versatile programming:

It can be programmed using AT commands for simple tasks, or with more advanced software like Arduino or the ESPRESSIF SDK for greater control.

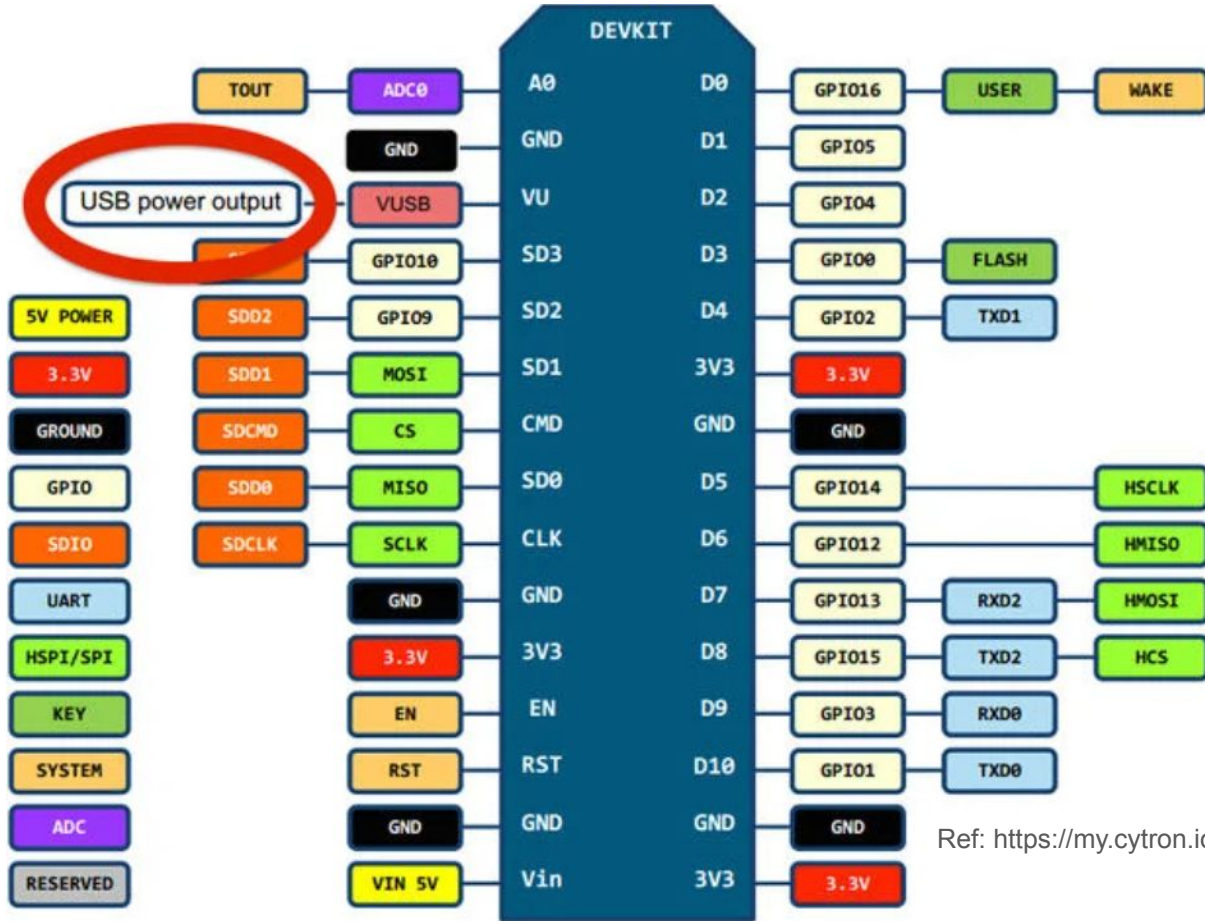
Large community support:

A significant and active community provides resources, documentation, and support for ESP8266 projects.

Multiple configurations:

Available in various forms, including modules like ESP-01, ESP-12, and NodeMCU, offering different features and pin layouts.

BASIC PINOUT & ITS CAPABILITIES



Ref: <https://my.cytron.io/p-nodemcu-lua-v3-esp8266-wifi-with-ch340c>

SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

Ref:

<https://www.instructables.com/Getting-Started-With-ESP8266LiLon-NodeMCU-V3Flashi/>

SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

Things you need:

[NodeMCU Flasher Master](#)

[NodeMCU Firmware](#)

[Arduino IDE](#)

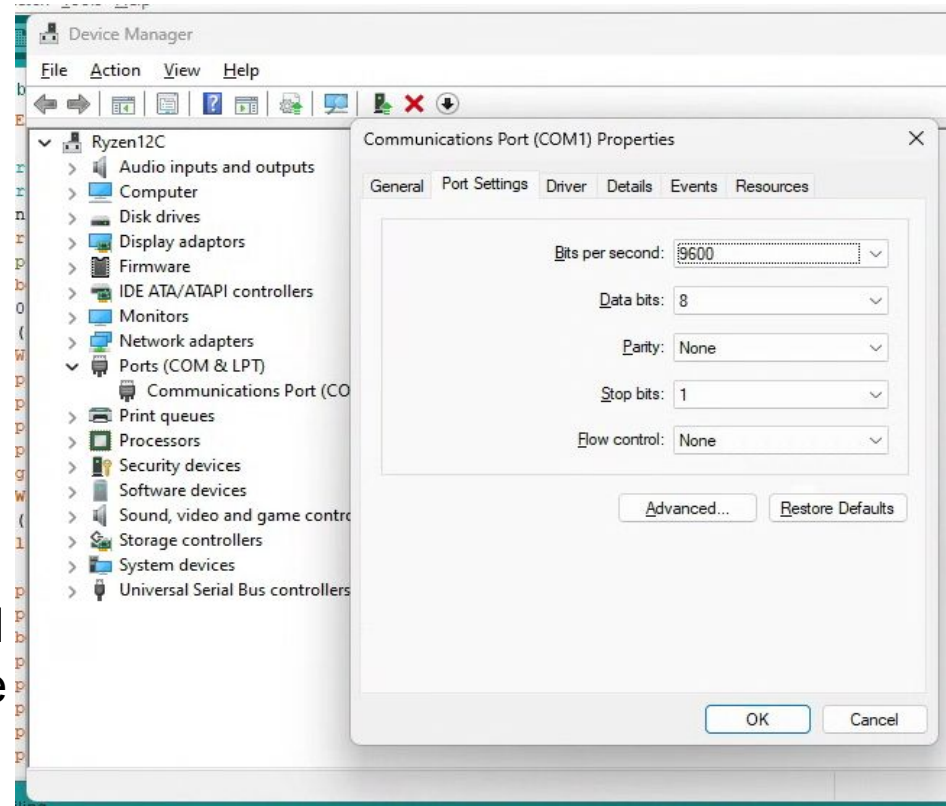
ESP8266MOD / NodeMCU V3

Micro USB Cable

[Reference](#)

SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

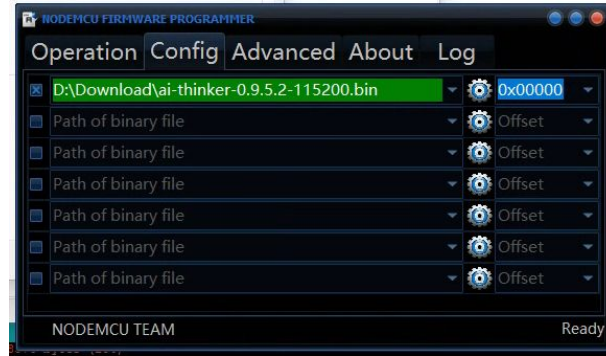
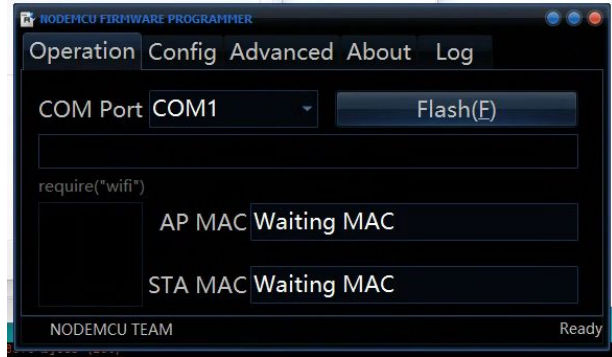
- Connect your ESP8266 to a laptop or PC using a micro USB cable.
- On a Windows system, search for and open '**Device Manager**'.
- Identify the COM port used to connect the ESP8266 and double-click it to open the COM Port Properties.
- Click on the '**Port Settings**' tab and change the '**Bits per second**' value to 115200.



SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

Installing the Firmware.

1. Extract NodeMCU flasher, and go to ..\nodemcu-flasher-master\Win64\Release and double click ESP8266Flasher. Select correct COM Port.



2. Click '**Config**' tab, click small gear icon and select the firmware XXX.bin

3. Click '**Advanced**' tab, change the Baudrate to 115200



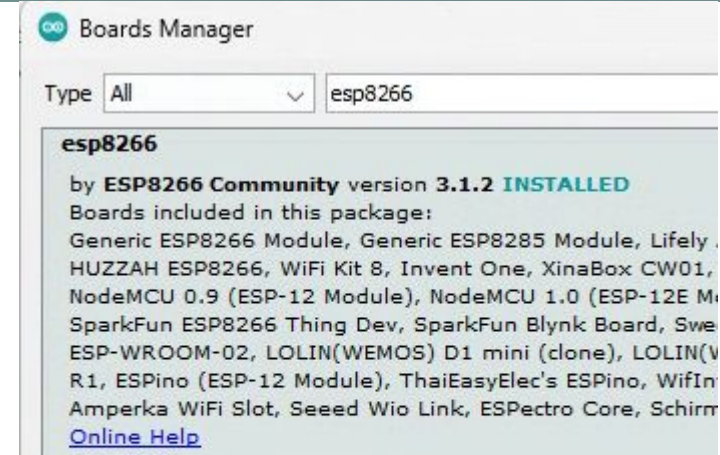
4. Click '**Operation**' tab again and click **Flash(F)** and wait the flashing process complete.



SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

Prep the Arduino IDE

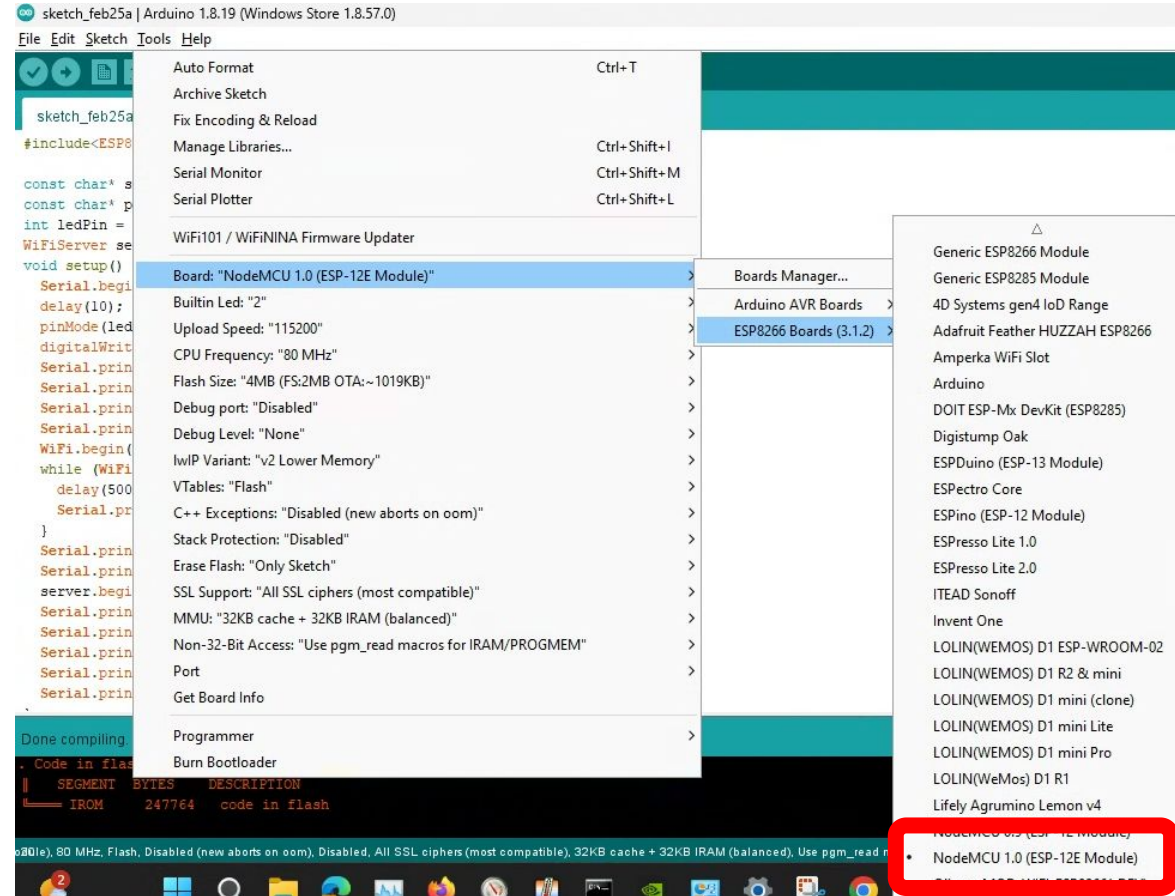
- Install the Arduino IDE.
- Open Arduino IDE.
- Click 'File' tab and open 'Preferences'.
- In the additional Boards Manager URLs add the following link (http://arduino.esp8266.com/stable/package_esp8266com_index.json) and click **OK**.
- Goto Tools > Boards > Boards Manager.
- Search for esp8266, click '**esp8266 by ESP8266 Community**' and click **Install**



SETTING UP THE ARDUINO IDE FOR ESP8266 DEV

Prep the Arduino IDE

- Goto Tools > Board > ESP8266 Boards and select NodeMCU 1.0 (ESP-12E Module) for our board.
- Goto Tools again, and select correct COM Port.



INTRODUCTION TO IOT & ITS TOOLS

Introduction to ThingSpeak

Ts. Dr. Razif bin Razali



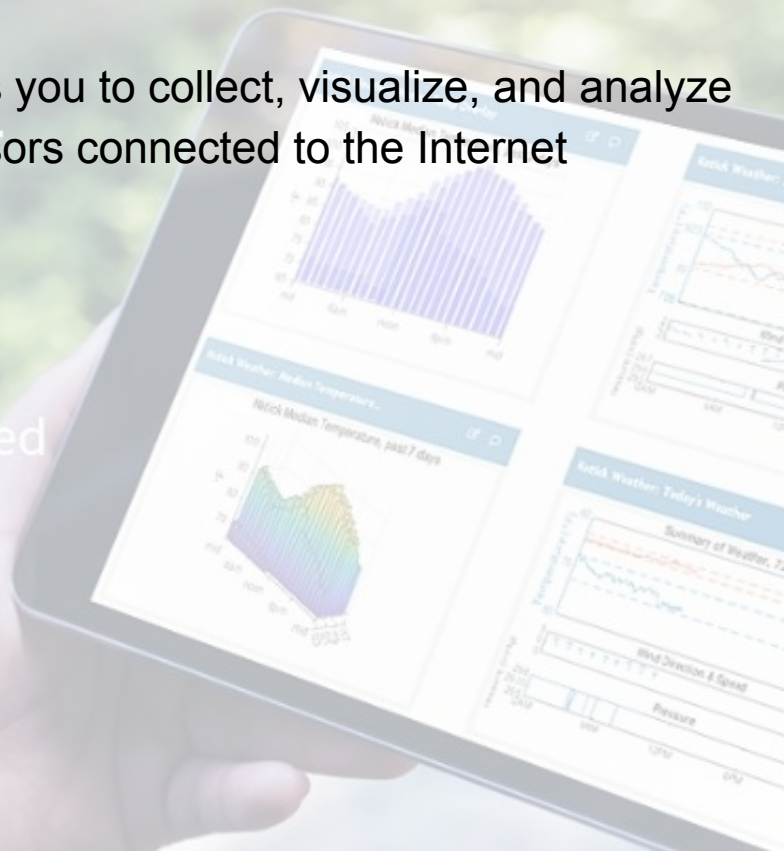
WHAT IS THINGSPEAK?

ThingSpeak is a cloud-based service that allows you to collect, visualize, and analyze live data streams from various devices and sensors connected to the Internet

ThingSpeak for IoT Projects

Data collection in the cloud with advanced
data analysis using MATLAB

ref : <https://thingspeak.mathworks.com/>



KEY FEATURES OF THINGSPEAK

Data Collection:

You can send data from your devices to ThingSpeak, which can then store it in private or public channels.

Visualization:

ThingSpeak provides instant visualizations of the data, allowing you to see trends and patterns in real-time.

Analytics:

You can use MATLAB analysis tools to process and analyze the data, including online analysis and processing.

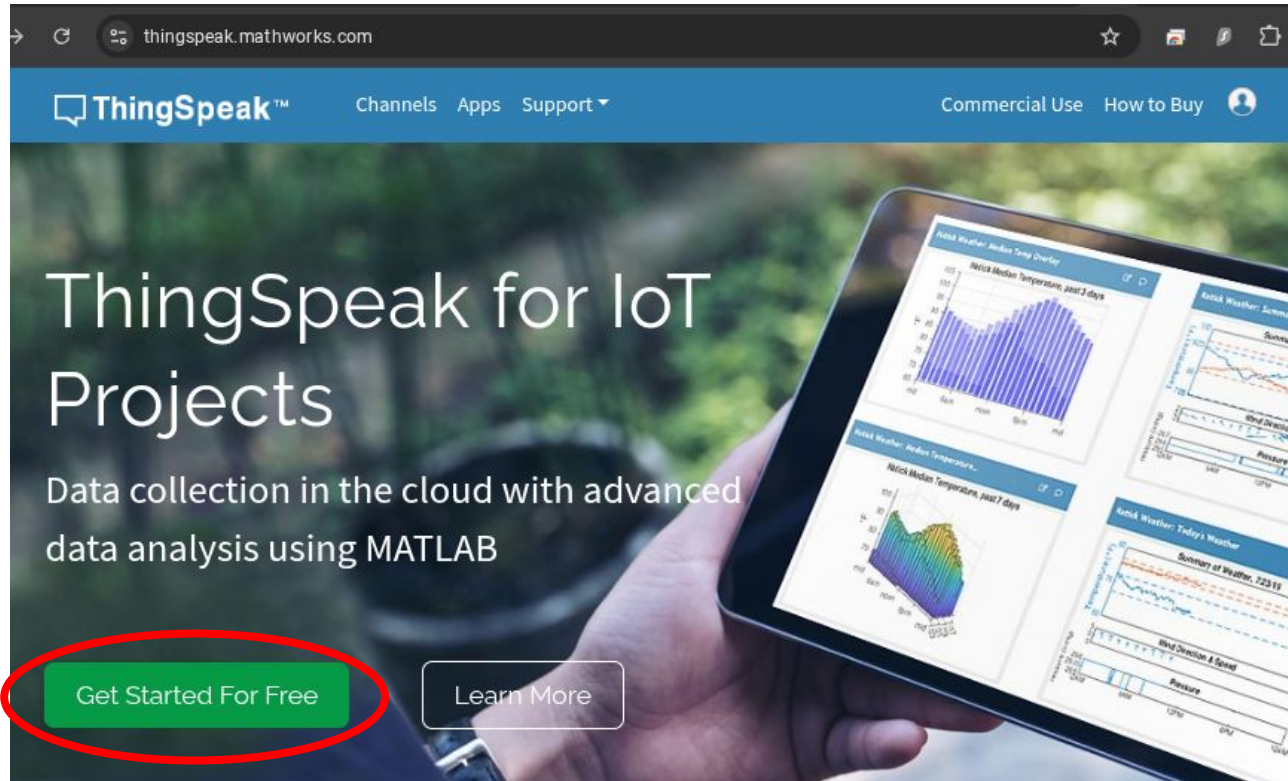
Actions:

ThingSpeak also allows you to set up alerts or triggers based on the data, so you can take action when certain conditions are met.

Integration:

You can integrate ThingSpeak with other web applications and systems using APIs.

CREATING THINGSPEAK ACCOUNT & NEW CHANNEL



CREATING THINGSPEAK ACCOUNT

To use ThingSpeak, you must sign in with your

Non-commercial users may use ThingSpeak for a time-limited free evaluation. To get full access, you need a university or organizational email address associated with your university or organization.

To send data faster to ThingSpeak or to send data from home and student usage.



Email

1

No account? [Create one!](#)
By signing in, you agree to our [privacy policy](#).

Create MathWorks Account

Email Address

Missing required information

To access your organization's MATLAB license, use your school or work email.

Location

First Name

Last Name

Continue

Personal Email Detected

To use your organization's MATLAB, enter your work or university email

Email Address

☒ Use this email for my MathWorks Account

Continue

This site is protected by reCAPTCHA and the Google [Privacy Policy](#) and [Terms of Service](#) apply.

Verify Your MathWorks Account

To finish creating your account, complete the following steps:

1. Go to your inbox for @gmail.com.
2. Click the link in the email we sent you.
3. Click **Continue**.

Didn't receive the email?

- Check your spam folder.
- [Send me the email again.](#)
- If you still have not received the email, Contact [Customer Support](#).

Continue

Verify Email Address

service@mail.account.mathworks.com
to me

8:18 PM (2 minutes ago)



Welcome to MathWorks!

To complete your MathWorks Account setup, click **Verify email**.

Verify email

Finish your Profile

Password

☐ I accept the [Online Services Agreement](#)

See our [privacy policy](#) for details.

Continue

Sign-up successful

Congratulations, you have successfully linked your MathWorks account to ThingSpeak. Use the following email ID and its associated MathWorks account password on all subsequent logins to ThingSpeak.

Email ID @gmail.com

Welcome to ThingSpeak!

OK

CREATE NEW CHANNEL

ThingSpeak™

My Channels

New Channel

Search by tag



Help

Collect data in a ThingSpeak channel from a device, from another channel, or from the web.

Click **New Channel** to create a new ThingSpeak channel.

Click on the column headers of the table to sort by the entries in that column or click on a tag to show channels with that tag.

Learn to [create channels](#), explore and transform data.

Learn more about [ThingSpeak Channels](#).

Examples

- [Arduino](#)
- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)
- [Netduino Plus](#)

Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

Upgrade

ThingSpeak™

New Channel

Name

TEMP & HUMIDITY

Description

RECORD TEMP AND HUMIDITY

Field 1

TEMP



Field 2

HUMIDITY



Field 3



Field 4



Video URL

http://

Show Status



Save Channel

Help

Channels store all the data that a ThingSpeak application collects. Each channel includes eight fields that can hold any type of data, plus three fields for location data and one for status data. Once you collect data in a channel, you can use ThingSpeak apps to analyze and visualize it.

Channel Settings

- **Percentage complete:** Calculated based on data entered into the various fields of a channel. Enter the name, description, location, URL, video, and tags to complete your channel.
- **Channel Name:** Enter a unique name for the ThingSpeak channel.
- **Description:** Enter a description of the ThingSpeak channel.
- **Field#:** Check the box to enable the field, and enter a field name. Each ThingSpeak channel has eight fields.

ThingSpeak™

TEMP & HUMIDITY

Channel ID: **2978383**

Author:

[mwa000037889969](#)

Access: Private

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

+ Add Visualizations

+ Add Widgets

Export recent data

MATLAB Analysis

MATLAB Visualization

Channel Stats

Created: [less than a minute ago](#)

Entries: 0

Field 1 Chart



TEMP & HUMIDITY

PRIVATE & PUBLIC VIEW OF THINGSPEAK



Channels ▾ Apps ▾ Devices ▾ Support ▾

Commercial Use How to Buy 

TEMP & HUMIDITY

Channel ID: 2978383

Author: mwa0000037889969

Access: Public

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

Channel Sharing Settings

- ☐ Keep channel view private
- ☒ Share channel view with everyone
- ☐ Share channel view only with the following users:

Email Address

Enter email here

Add User

Help

ThingSpeak allows you to control who can view the data in your channel. Irrespective of the settings on this tab, reading data from or writing data to the fields of a channel requires the appropriate API key for the channel.

Channel Sharing Settings

- **Keep channel view private:** Selecting this option keeps your channel private. Only you will be able to see the channel view.
- **Share channel view with everyone:** Selecting this option makes the public view of your channel viewable by anyone browsing the ThingSpeak website.
- **Share channel view only with the following users:** Selecting this option shares the private view of your channel only with specific ThingSpeak users.

UNDERSTANDING CHANNEL ID & API KEYS

Channel Information

- **TEMP & HUMIDITY:** This is the **Name** of the current ThingSpeak Channel. It's user-defined and describes the type of data being collected.
- **Channel ID: 2978383:** This is a unique numerical identifier for this specific channel. You'll use this ID in your device code or API requests to specify which channel to interact with.
- **Author: mwA0000037889969:** The username or identifier of the ThingSpeak user who created or owns this channel.
- **Access: Public:** This indicates that the data in this channel (specifically the "Public View") can be viewed by anyone without needing an API key. If it were "Private," only the owner or those with a Read API Key could see the data.
- **RECORD TEMP AND HUMIDITY:** This appears to be a **Description** for the channel, providing more context about its purpose.

ThingSpeak™

Channels ▾ Apps ▾ Devices ▾ Support ▾

Commercial Use How to Buy WA

TEMP & HUMIDITY

Channel ID: 2978383
Author: mwA0000037889969
Access: Public

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

Write API Key

KeyCUPRHU8PH0VRD6SM

Generate New Write API Key

Read API Keys

KeyYB1ECI9QJBJOAYH

Note

Save NoteDelete API Key

Add New Read API Key

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- **Write API Key:** Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key**.
- **Read API Keys:** Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- **Note:** Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api_key=CUPRHU8PH0VRD6SM&field1=0

Read a Channel Feed

GET https://api.thingspeak.com/channels/2978383/feeds.json?results=2

Read a Channel Field

GET https://api.thingspeak.com/channels/2978383/fields/1.json?results=2

Read Channel Status Updates

GET https://api.thingspeak.com/channels/2978383/status.json

Learn More

UNDERSTANDING CHANNEL ID & API KEYS

Write API Key:

- **Key: CUPRHU8PH0VRD6SM:** This is a secret alphanumeric string. **This key allows you or your IoT device to send data to this ThingSpeak channel.** It's like a password for writing data. You embed this key in the code of your sensor device (e.g., an ESP32, Arduino) so it can update the channel fields.

Read API Keys:

- **Key: YB1ECI9QJBJOAOYH:** This is also a secret alphanumeric string. **This key allows you or applications to read data from this ThingSpeak channel, especially if the channel is private.** If the channel is public, you might not always need a Read API key to fetch data,

ThingSpeak™

Channels ▾ Apps ▾ Devices ▾ Support ▾

Commercial Use How to Buy WA

TEMP & HUMIDITY

Channel ID: 2978383
Author: mwa000037889969
Access: Public

RECORD TEMP AND HUMIDITY

Private View Public View Channel Settings Sharing API Keys Data Import / Export

Write API Key

Key CUPRHU8PH0VRD6SM

Generate New Write API Key

Read API Keys

Key YB1ECI9QJBJOAOYH

Note

Save Note Delete API Key

Add New Read API Key

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API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api_key=CUPRHU8PH0VRD6SM&field1=0

Read a Channel Feed

GET <https://api.thingspeak.com/channels/2978383/feeds.json?results=2>

Read a Channel Field

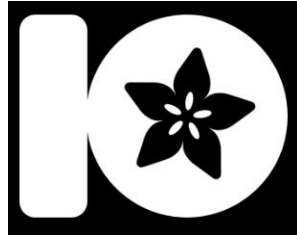
GET <https://api.thingspeak.com/channels/2978383/fields/1.json?results=2>

Read Channel Status Updates

GET <https://api.thingspeak.com/channels/2978383/status.json>

Learn More

OTHER THAN THINGSPEAK



EMQX